

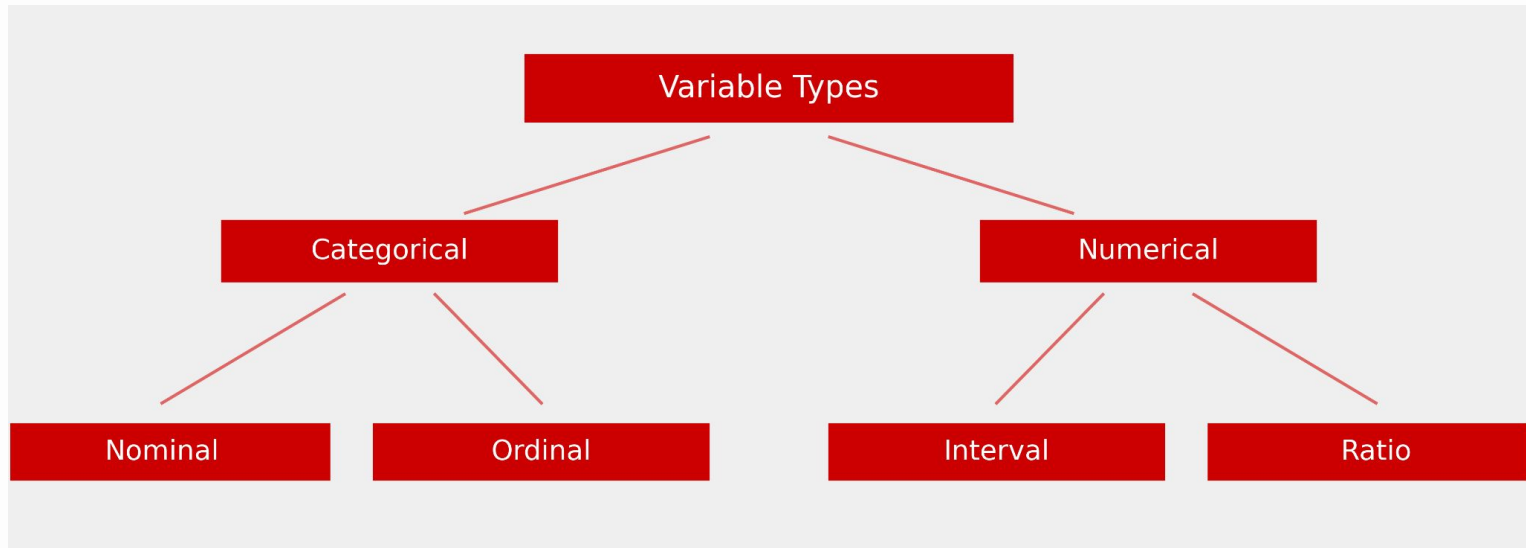
Data Science

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[BASIC] STATISTICS

Types of Variables

- There are many taxonomies for variables.
 - In this module, we will follow a simple, yet robust categorization:



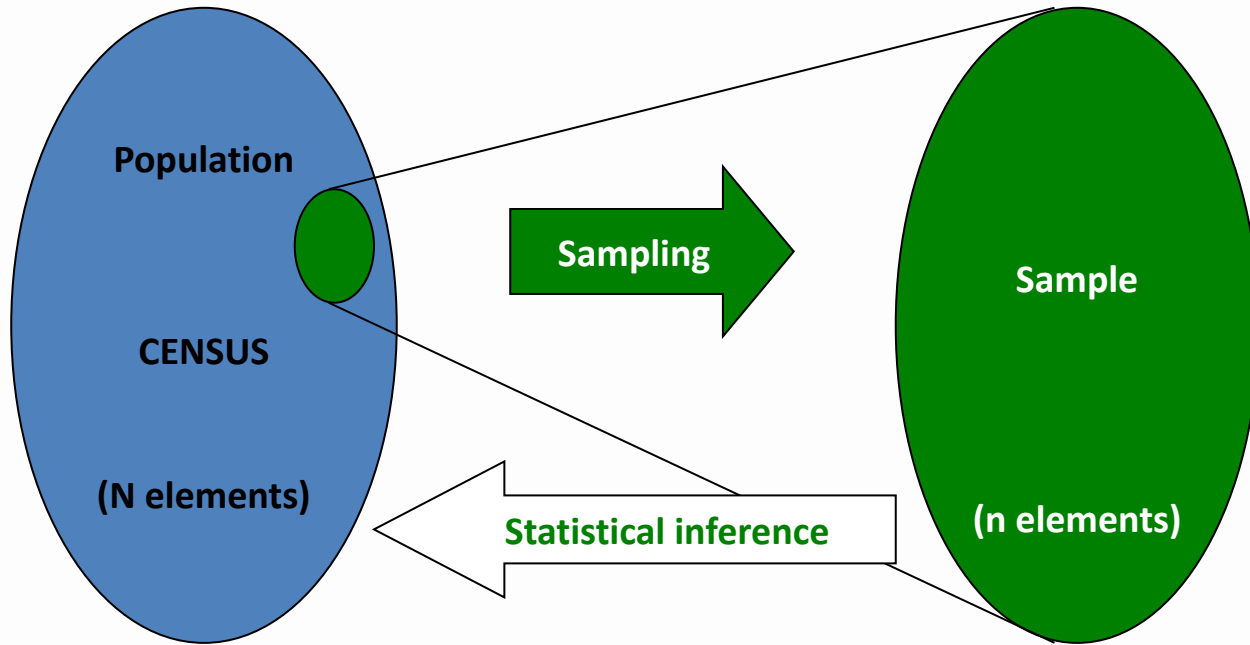
Types of Variables

- **Categorical**
 - Nominal
 - Colors (red, blue, green) and gender (male, female);
 - All values share the same importance.
 - Ordinal
 - Educational attainment (e.g., Elementary school, High school, Bachelor's, Master's, PhD) and satisfaction ratings (poor, average, good, excellent)

Types of Variables

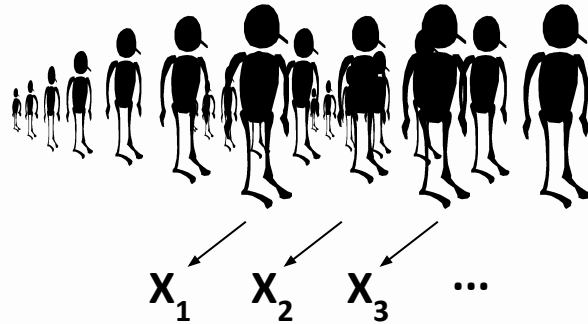
- **Numeric**
 - Interval
 - These variables have ordered values, but no true zero point
 - Monetary amounts, temperature in Celsius or Fahrenheit, IQ values, etc
 - Ratio
 - Same as above, but the 0 (zero) value has a “special meaning” (true zero)
 - Height, weight, temperature in Kelvin, and income (none of which can be negative – and some, in fact, cannot be zero!)

Population vs. Sample



Population vs. Sample

- **Population**: the complete set of individuals that share at least one observable characteristic.



- If data is collected from **every individual** in the population → **Census**
- If data is collected from **only a subset** of the population → **Sample**

Sampling

- Process of selecting a **subset of elements** from a larger population
 - If the sample is **representative**, we can use it to **estimate characteristics of the population**.

Why do we sample?

Sampling

- Process of selecting a **subset of elements** from a larger population
 - If the sample is **representative**, we can use it to **estimate characteristics of the population**.

Why do we sample?

- The population may be very large or even infinite!
- Collecting data from everyone can be expensive and time-consuming
 - Sampling reduces data collection costs and effort

How do we sample?

- The sample size depends on:
 - data variability (greater variance $\Rightarrow$ greater n)
 - of the precision required (greater precision $\Rightarrow$ greater n)
 - the time available (less time $\Rightarrow$ smaller n)
 - the sampling cost (higher cost $\Rightarrow$ smaller n)

Sample Representativity

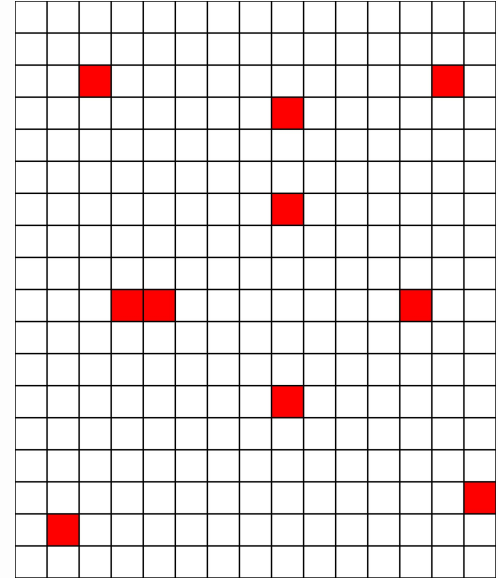
- **Sample randomness**
 - The key is to guarantee that every individual from the population has a chance of participating of the sample
- **Unbiased** sampling
 - The selection process should **not systematically favor certain individuals**
- The only way to generalize the results from the sample to the entire population

Sampling Methods

- PROBABILISTIC
 - Each population member has the same (non-null) probability of being selected to the sample
- NON PROBABILISTIC
 - Sampling is restricted to elements that we have access to (e.g., chemical dependants in treatment)
 - Impossibility of sampling (e.g., blood test)
 - Volunteering (e.g., vaccine and/or medication testing)

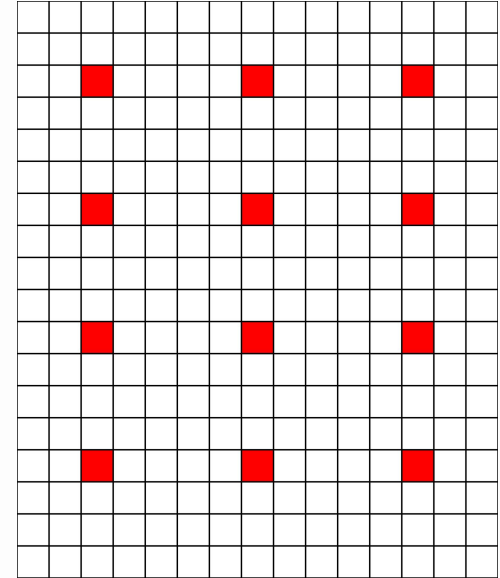
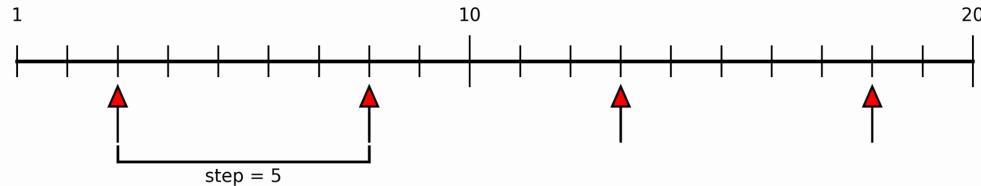
Simple Random Sampling

- A sample of **size n** is randomly selected from a **population of size N**
- Each element has the **same probability** of being selected
- Sampling can be done **with or without replacement**



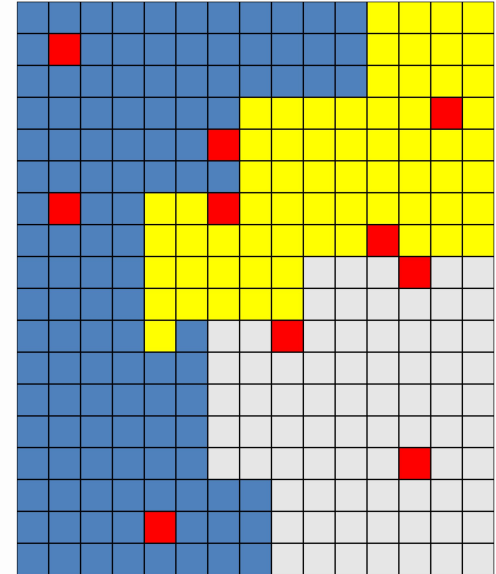
Systematic Sampling

- Sampling is performed **uniformly across the population**
 - If the population elements are **ordered according to some criterion**, samples can be selected using a fixed **step (interval)**



Stratified Sampling

- Produces **more precise samples** by representing different groups in the population
- Suitable for **heterogeneous populations with homogeneous strata**
- The population (**N**) is divided into **L strata** with sizes $N_1, N_2, \dots, N_L$
 - In each stratum, elements are randomly selected



Reminder

- Most of the datasets we will work with are samples that represent behaviors we observe in the real world
- Our goal in machine learning is to learn patterns from these **samples** and build models that **generalize well**
 - A good model should be able to accurately predict the behavior of new, **unseen examples**, not just the data used during training

Descriptive Statistics

- Goal: summarize and describe a dataset
- Main statistics:
 - Mean (mean)
 - Mode (mode)
 - Median (median)
 - Quartiles and Percentiles (quantile)
 - Variance (var)
 - Standard deviation (std)

Mean

- Computed as the **sum of all values divided by the number of observations**:

$$\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i$$

- It is the **most widely used statistic** for summarizing numerical data.
 - The mean does not necessarily correspond to a value present in the dataset!

Mode

- The **most frequent value** in a dataset
- A dataset may have **one mode, multiple modes, or no mode**
- What is the **mode** in the following lists?
 - [1, 1, 2, 3, 4]
 - [1, 2, 3, 4, 5, 6, 7, 8, 9, 10]
 - [1, 1, 1, 2, 2, 2, 3, 4, 5]

Mode

- The **most frequent value** in a dataset
- A dataset may have **one mode, multiple modes, or no mode**
- What is the **mode** in the following lists?
 - $[1, 1, 2, 3, 4] \rightarrow 1$ (appears most frequently)
 - $[1, 2, 3, 4, 5, 6, 7, 8, 9, 10] \rightarrow \text{No mode}$ (all values appear once)
 - $[1, 1, 1, 2, 2, 2, 3, 4, 5]$
 - **1 and 2** (both appear 3 times $\rightarrow$ bimodal distribution)

Median

- The **middle value** of a dataset after the values are sorted.
- Example:
 - Original: [32, 33, 24, 31, 44, 65, 32, 21, 32]

Median

- The **middle value** of a dataset after the values are sorted.
- Example:
 - Original: [32, 33, 24, 31, 44, 65, 32, 21, 32]
 - Sorted: [21, 24, 31, 32, 32, 32, 33, 44, 65]

Median

- The **middle value** of a dataset after the values are sorted.
- Example:
 - Original: [32, 33, 24, 31, 44, 65, 32, 21, 32]
 - Sorted: [21, 24, 31, 32, 32, 32, 33, 44, 65]
 - Median: [21, 24, 31, 32, **32**, 32, 33, 44, 65]

Median

- What if a dataset has an **even number of elements**?
- Example:
 - [18, 19, 19, 22, 44, 45, 46, 46, 47, 48]

Median

- If a dataset has an **even number of elements**, the median is typically computed as the **average of the two central values**
- Example:
 - [18, 19, 19, 22, **44**, **45**, 46, 46, 47, 48]
 - Median = (**44** + **45**) / 2 = **44,5**

Quartiles

- Quartiles divide a **sorted dataset** into **four equal parts**.
 - They are denoted by **Q1**, **Q2**, and **Q3**, where **Q2** corresponds to the median:

[18, 19, 19, 22, 44, 45, 46, 46, 47, 48]

Q2 = 44,5
(median)

Q1 = ?

Q3 = ?

Quartiles

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$$Q1 = 19$$

$$Q3 = 46,5$$

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[18, 19, 19, 22, 44, 45, 46, 46, 47, 48]

$$Q2 = 44,5$$

(median)

Interpretation:

- **Q1 = 19** $\rightarrow$ 25% of the data are ≤ 19
- **Q2 = 44,5** $\rightarrow$ 50% of the data are $\leq 44,5$
- **Q3 = 46,5** $\rightarrow$ 75% of the data are $\leq 46,5$

Deciles and Percentiles

- Deciles and Percentiles extend the same idea
- **Deciles** divide a dataset into **10 equal parts**
 - They are represented as D1, D2, D3, ..., D9
 - **D7** → 70% of the observations are below or equal to that value
- **Percentiles** divide the data into **100 equal parts**
 - They are represented as P1, P2, P3, ..., P99
 - **P90** → 90% of the observations are below or equal to that value

Variance

- Measures how much the values in a dataset **differ from the mean**
 - A **smaller variance** indicates that the values are **closer to the mean**, while a **larger variance** indicates **greater dispersion**
- The sample variance is given by:
$$Var(X) = \frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n - 1}$$
- Where:
 - $\bar{x}$ is the **mean of the dataset**
 - n is the **number of observations (sample size)**
 - $\bar{x}_i$ represents each **individual value** in the dataset

Why Don't We Usually Report Variance?

Variance measures dispersion, but it has a major **limitation**:

- ???

Why Don't We Usually Report Variance?

Variance measures dispersion, but it has a major **limitation**:

- Variance is expressed in **squared units**
 - **Example**: if the data is measured in **meters**, variance is measured in **meters²**
 - **This makes interpretation difficult!**

Example:

- Mean height = 1.75 m
- Variance = 0.04 m²
 - What does 0.04 m² mean in practice?

Standard Deviation (std dev)

- The standard deviation is the **square root of the variance**

$$\sigma = \sqrt{Var(X)}$$

- The **standard deviation** measures how much the values in a dataset **vary around the mean**
 - It can be interpreted as the error when replacing individual values with the mean of the dataset
- Standard deviation (σ) is usually **reported together with the mean** to summarize the data:

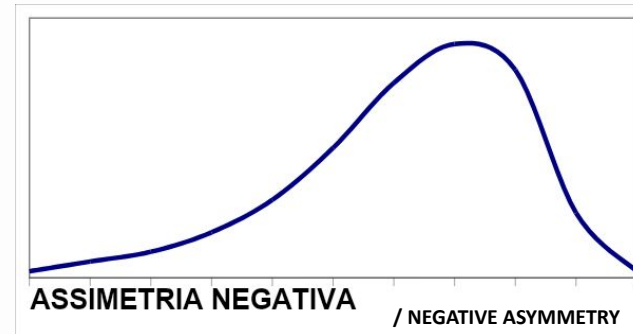
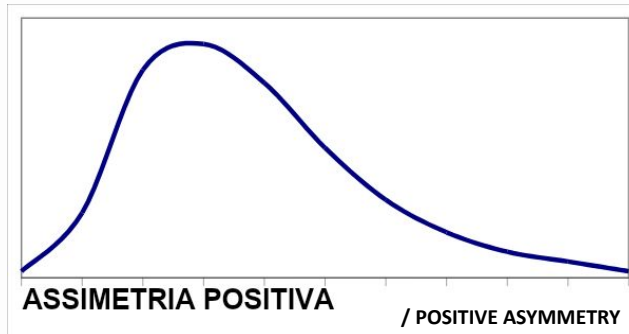
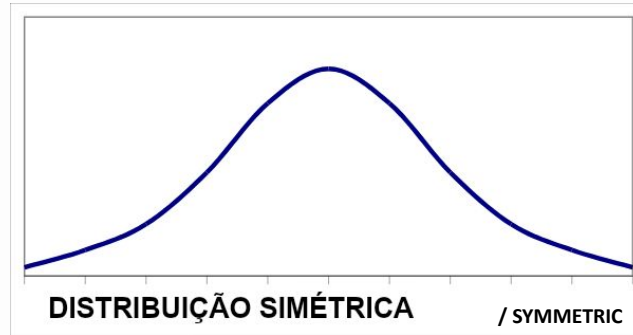
$$\bar{x} \pm \sigma$$

ANALYZING DISTRIBUTIONS

Skewness (Asymmetry)

- Measures the **asymmetry** of a distribution relative to a **Gaussian (normal) distribution**
 - When the **mean, median, and mode are equal**, the distribution is **symmetric**, and the **skewness coefficient is zero**
 - When the **mean is greater** than the median and mode, the distribution has **positive skewness** (right-skewed)
 - When the **mean is smaller** than the median and mode, the distribution has **negative skewness** (left-skewed)

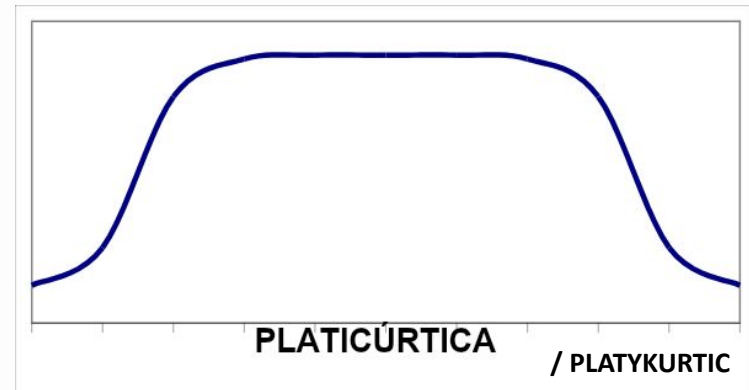
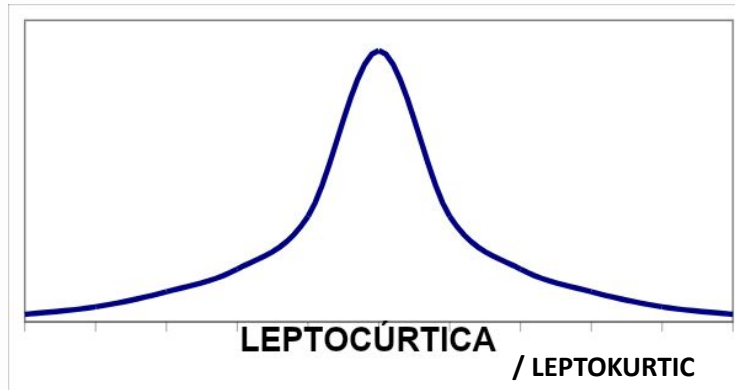
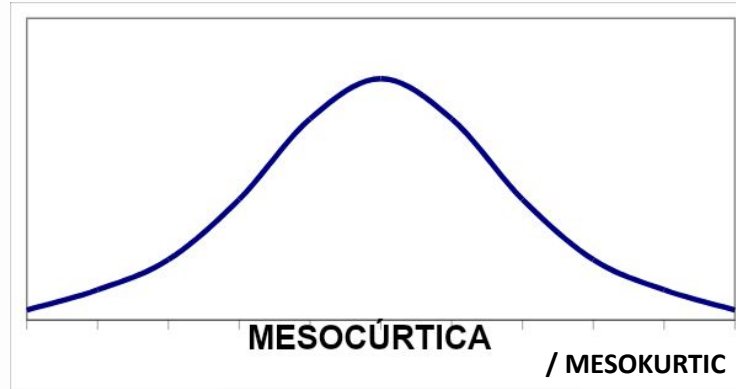
Skewness (Asymmetry)

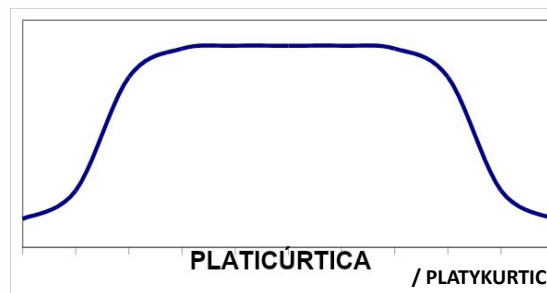
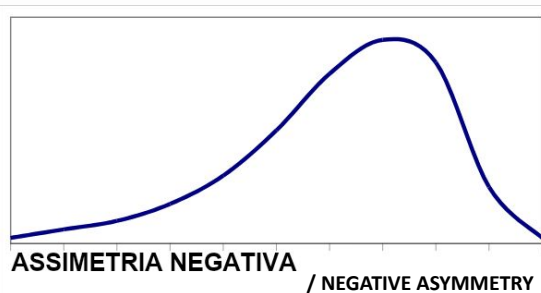
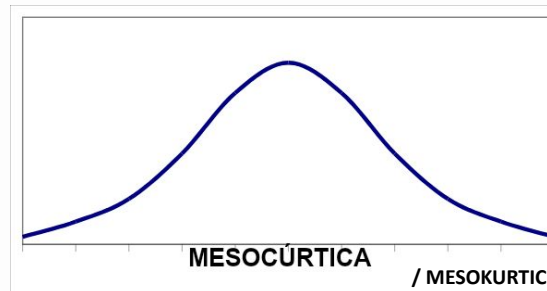
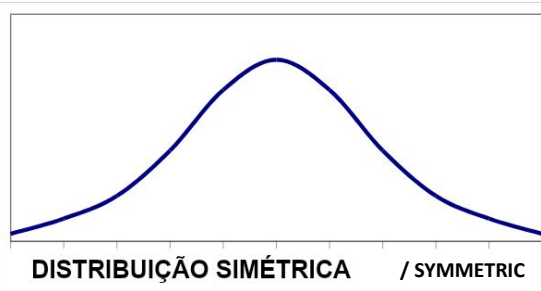
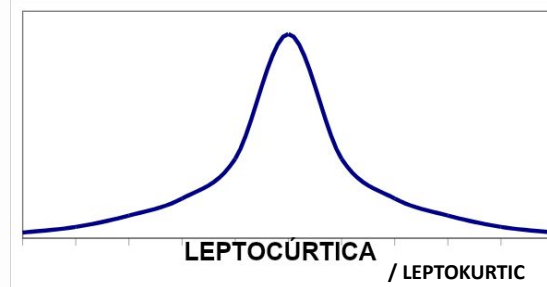
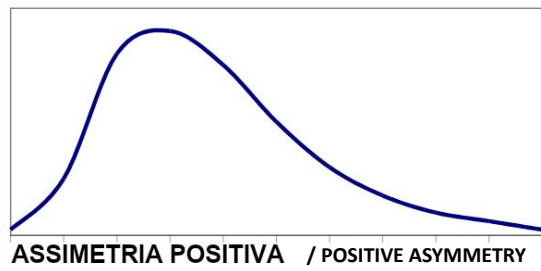


Kurtosis

- Measures the “tailedness” of a distribution compared to a Gaussian (normal) distribution
 - A Gaussian distribution has **zero kurtosis** and is called **mesokurtic**
 - A distribution with **positive kurtosis** is called **leptokurtic** (heavier tails and more extreme values)
 - A distribution with **negative kurtosis** is called **platykurtic** (lighter tails and fewer extreme values)

Kurtosis





Why is this important?

- Many machine learning algorithms assume that the data follows a **Gaussian (normal) distribution**
 - When the data is highly skewed, these assumptions may no longer hold!
- Therefore, it is often necessary to transform or preprocess the data to address such behavior
 - We will explore different techniques to handle this 😊

Practice

Practice #1 – Recap from Last Class

- Using the [salaries.csv](#) dataset, compute the following:
 - What is the **average salary of professors across different disciplines?**
 - What are the **average salary** and the **number of male professors in Discipline B** who completed their PhD in **less than 5 years?**
 - What is the **average salary of professors with less than 5 years of teaching experience?** And how does it **compare to those with more than 40 years of experience?**

Practice #2

- Pick a number between **1 and 12**;
- Download the dataset and replace **X** with your chosen number:
 - `df = pd.read_csv('https://raysonlaroca.github.io/teaching/data-science/datasets/ans/datasetX.csv')`
- Use your pandas skills to analyze the dataset and figure out what is happening in the data.

Anscombe Quartet

- **Same mean, variance, correlation and linear regression coefficients!**
 - Therefore, summary statistics alone are not sufficient;
 - **Visualization is extremely important**, as the underlying structure and behavior of the data can be very different;
 - See the original paper:
 - F.J. Anscombe, "[Graphs in Statistical Analysis](#)," American Statistician, 27 (February 1973), 17-21.

Same Stats, Different Graphs

Same Stats, Different Graphs:

Generating Datasets with Varied Appearance and Identical Statistics through Simulated Annealing